



MULTIMEDIA SIGNAL PROCESSING

LAB MANUAL

VI Semester (22ETL642)



Name of the Student :

Semester /Section :

USN :

Batch :

Vision of the Institute

To impart quality technical education with a focus on Research and Innovation emphasizing on Development of Sustainable and Inclusive Technology for the benefit of society.

Mission of the Institute

- To provide an environment that enhances creativity and Innovation in pursuit of Excellence.
- To nurture teamwork in order to transform individuals as responsible leaders and entrepreneurs.
- To train the students to the changing technical scenario and make them to understand the importance of Sustainable and Inclusive technologies.



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DAYANANDA SAGAR COLLEGE OF ENGINEERING
(An Autonomous Institution affiliated to Visvesvaraya Technological University, Belagavi)
DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING,
BENGALURU-560078

DAYANANDA SAGAR COLLEGE OF ENGINEERING
(An Autonomous Institution affiliated to Visvesvaraya Technological University, Belagavi)
ELECTRONICS AND TELECOMMUNICATION ENGINEERING
DEPARTMENT
SHAVIGE MALLESWARA HILLS
KUMARASWAMY LAYOUT
BENGALURU-560078

VISION OF THE DEPARTMENT

➤ *To disseminate quality technical education for achieving Academic Excellence through focused research encompassing Mobile, Sensor, and Telecommunication networks with a thrust on Space Communication and Telecommunication Standards.*

MISSION OF THE DEPARTMENT

- *By disseminating the knowledge of devices, systems, and technologies that are impacting the Telecommunication field.*
- *By educating students towards multidisciplinary practices of Telecommunication industry and standards for a successful career.*
- *By inculcating research attitude among graduates through continuous life long learning related to Telecommunication and its allied domain*

PROGRAMME EDUCATIONAL OBJECTIVES [PEOs]

Program Educational Objective-1 (PEO1): Graduate contributes service to Telecommunication Engineering industry, government organizations, and society by applying skills acquired through formal technical education.

Program Educational Objective-2 (PEO2): The graduate demonstrates ability to apply Scientific, Mathematical and Telecommunication Engineering knowledge for solving real life problems.

Program Educational Objective-3 (PEO3): The graduate possess temperament of professional attitude, effective communication skills, and multidisciplinary approach to resolve Telecommunication domain specific challenges.

Program Educational Objective-4 (PEO4): The graduate demonstrates continuous learning to update knowledge of emerging Telecommunication technologies through higher education, for sustainable career development to

strengthen human values, with a focus on safety and environmental concern.

PROGRAMME SPECIFIC OUTCOMES [PSOs]

PSO – 1: To analyze, design and implement Engineering solutions for Signal Processing application and Communication Systems.

PSO – 2 : To design, synthesize and provide solutions for Communication Network Applications

PSO – 3: To analyze design and synthesize Embedded based System for real time scenarios.

**DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING,
BENGALURU-560078**

**Multimedia Signal Processing lab
(SYLLABUS)**

VI SEMESTER B. E (ET)

Sub. Code:
Hrs/Week: 3
Total Hrs:40

IA Marks : 25
Exams Hrs : 3
Exam Marks: 50

Course Objectives:

- Acquire the essentials for comprehending the basic concepts of various multimedia networks and applications.
- Recognize the basic principles of digitization and the techniques needed to analyze various media formats.
- Explore the processing and compression methods needed to compress the size of multimedia data.

Course Outcomes:

Cos	Course Outcomes
CO1	Apply various fundamental concepts and analyse performance of multimedia Signal processing
CO2	Performance analyze of various file formats of audio, video and text media in different applications
CO3	Implement and evaluate different types of multimedia compression methods
CO4	Ability to work effectively in a team to carry out experiments and evaluate for an identified multimedia application or problem statement applying the modern engineering tool.

Syllabus:

Part A

List of Experiments	
Part-A(cycle-1)	
1	To study the Multimedia concept.
2	a. To obtain histogram equalization of an image. b. To implement smoothing or averaging filter in spatial domain.
3	a. Program for opening and closing of the image. b. To fill the region of interest for the image. c. Program for edge detection algorithm.
4	a. Program of sharpen image using gradient mask. b. Program for morphological operation: erosion and dilation.
5	Program for DCT/IDCT computation
PART-B(cycle-2)	
6	Program to read the video and displaying the frames of the video
7	Program to process and write the processed video
8	Program to read, write and process an Audio file
9	Program to add noise to audio file and remove noise using filter

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DO's

- Adhere and follow timings, proper dress code with appropriate foot wear.
- Bags and other personal items must be stored in designated place.
- Come prepared with the viva, procedure, and other details of the experiment.
- Secure long hair, Avoid-loose clothing , Deep neck and sleeveless dresses
- Do check for the correct ranges/rating and carry one meter/instrument at a time
- Inspect all equipment/meters for damage prior to use
- Conduct the experiments accurately as directed by the teacher.
- Immediately report any sparks/ accidents/ injuries/ any other troublesome incident to the faculty /instructor.
- Handle the apparatus/meters/computers gently and with care
- In case of an emergency or accident, follow the safety procedure.
- Switch OFF the power supply after completion of experiment

DONT's

- The use of mobile/ any other personal electronic gadgets is prohibited in the laboratory.
- Do not make noise in the Laboratory & do not sit on experiment table.
- Do not make loose connections and avoid overlapping of wires
- Don't switch on power supply without prior permission from the concerned staff.
- Never point/touch the CRO/Monitor screen with the tip of the open pen/pencil/any other sharp object.
- Never leave the experiments while in progress.
- Do not insert/use pen drive/any other storage devices into the CPU?
- Do not leave the Laboratory without the signature of the concerned staff in observation book

Experiment No: 1

Date: _____

AIM: To study the Image Processing concept.

OBJECTIVE: To study the Image Processing concept.

TOOLS REQUIRED: MATLAB

THEORY: Digital images play an important role both in daily life applications as well as in the areas of research technology. The digital image processing refers to the manipulation of an image by means of processor. The different elements of an image processing system include image acquisition, image storage, image processing and display

An image is two dimensional function that represent a message of sum characteristics such as brightness or color of viewed scene in the first mat lab program the command used from mat lab is imcomplement

PROGRAM:

% Program to study the image processing concept

```
I=imread('pout.tif');
```

```
J=imcomplement(I);
```

```
figure,imshow(I)
```

```
figure,imshow(J)
```

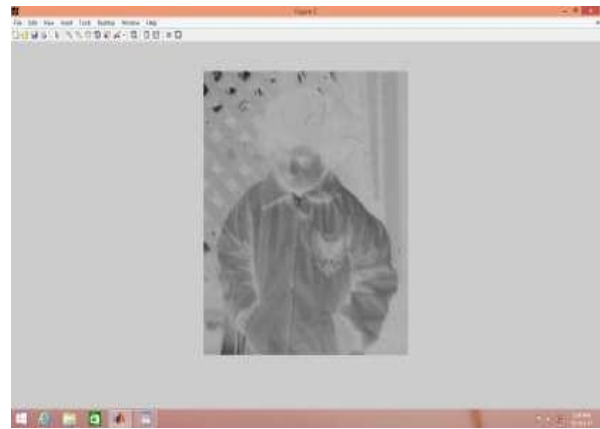
```
K=imadjust(I,[0;0.4],[0.5;1])
```

```
figure,imshow(K)
```

Result:



Original Image



Complement Image

Conclusion: Thus we have studied the how to obtain complement image from the original image.

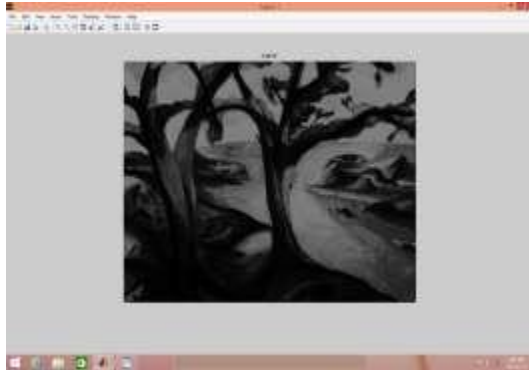
Experiment No: 2 Date: **2a): AIM:** To obtain histogram equalization image.**OBJECTIVE:** To obtain histogram equalization image.**TOOLS REQUIRED:** MATLAB

THEORY: Histogram equalization is a method in image processing of contrast adjustment using the image's histogram. Histogram equalization often produces unrealistic effects in photographs; however it is very useful for scientific images like thermal or x-ray images, often the same class of images to which one would apply false color. Also histogram equalization can produce undesirable effects when applied to images with low color depth. For example, if applied to 8-bit image displayed with 8 bit gray scale it will further reduce color depth (number of unique shades of gray) of the image. Histogram equalization will work the best when applied to images with much higher depth than palette size, like continuous data or 16-bit gray-scale images.

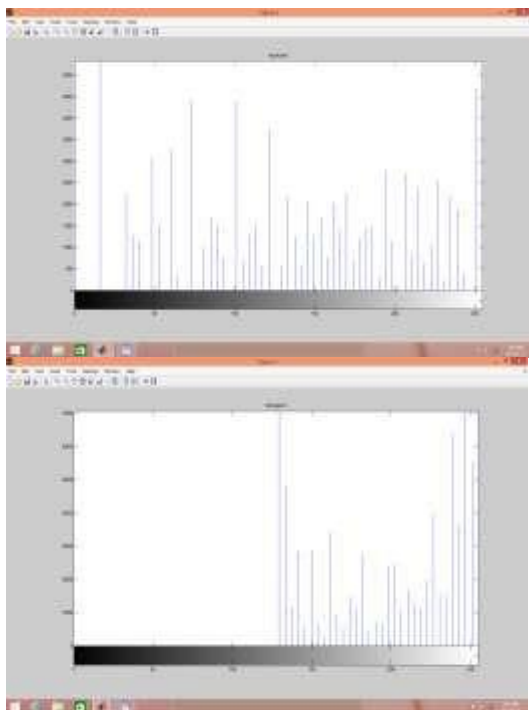
% Program to obtain histogram equalization concept

```
I=imread('trees.tif');  
J=imcomplement(I);  
imhist(J,100); imshow(I);  
title('original');  
figure,imshow(J);  
title('complement');  
I=histeq(I);  
figure,imhist(I,64);  
title('equilized');  
figure,imhist(J,64);  
title('histogram');  
n=numel(I); p=imhist(I)/n;  
figure,plot(p);  
title('normalized');  
K=imadjust(I,[0;1],[0.4;1],0.5);  
figure,imshow(K);  
title('adjusted image');  
T=maketform('affine',[.3 0 0;.5 1 0;0 1 1]);  
tformfwd([0,0],T);  
I2=imtransform(I,T);  
figure,imshow(I2);  
title('forward image');
```

Result:



Original Histogram



Equalized Histogram

2b): AIM: To Implement smoothing or averaging filter in spatial domain.

OBJECTIVE: To Implement smoothing or averaging filter in spatial domain.

TOOLS REQUIRED: MATLAB

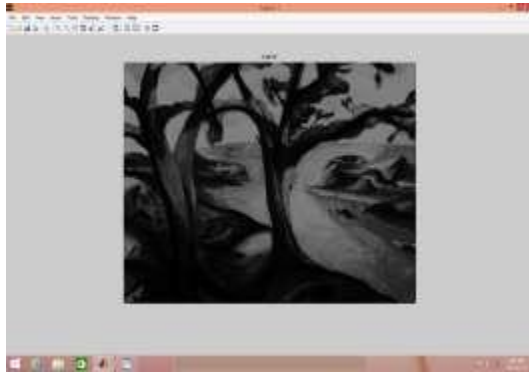
THEORY: Filtering is a technique for modifying or enhancing an image.....Mask or filters will be defined. The general process of convolution and correlation will be introduced via an example. Also smoothing linear filters such as box and weighted average filters will be introduced.

In statistic and image processing, to smooth a data set is to create an approximating function that attempts to capture important patterns in the data, while leaving out noise or other fine-scale structures/rapid phenomena. In smoothing, the data points of a signal are modified so individual points (presumably because of noise) are reduced, and points that are lower than the adjacent points are increased leading to a smoother signal. Smoothing may be used in two important ways that can aid in data analysis by being able to extract more information from the data as long as the assumption of smoothing is reasonable by being able to provide analyses that are both flexible and robust, different algorithms are used in smoothing.

PROGRAM:

```
% Program for implementation of smoothing or averaging filter in spatial domain
I=imread('trees.tif');
subplot(2,2,1);
imshow(I);
title('original image');
f=ones(3,3)/9;
h=imfilter(I,f,'circular');
subplot(2,2,2);
imshow(h);
title('averaged image');
```

Result:



Conclusion: Thus we have performed the smoothing or averaging filter operation on the Original image and we get filtered image.

Experiment No: 3

Date: _____

3a): AIM: Program for opening and closing of the image.**OBJECTIVE:** Program for opening and closing of the image.**TOOLS REQUIRED: MATLAB**

THEORY: In mathematical morphology, opening is the dilation of the erosion of a set by a structuring elements B:

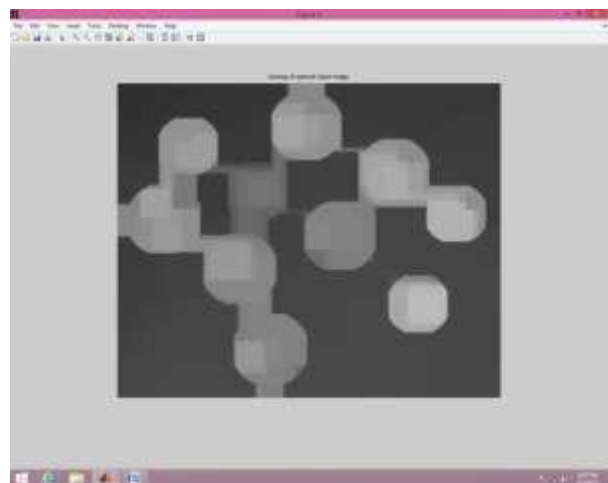
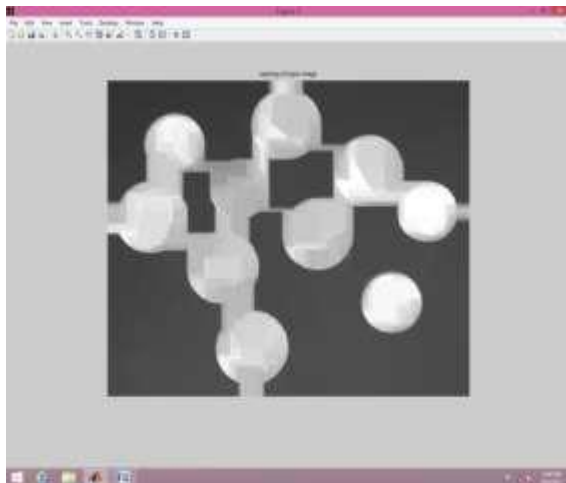
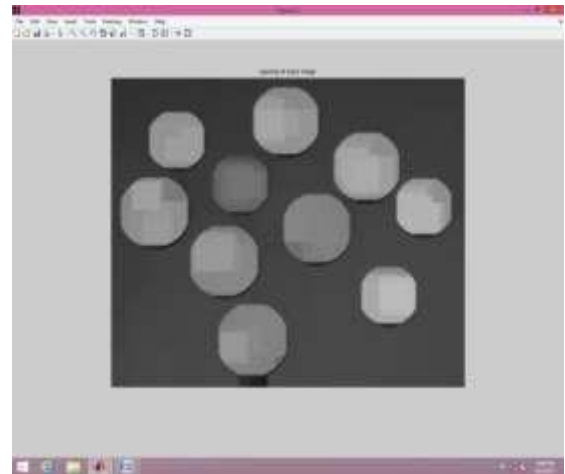
Together with closing, the opening serves in computer vision and image processing as a basic workhorse of morphological noise removal. Opening removes small objects from the foreground (usually taken as the bright pixels) of an image, placing them in the background, while closing removes small holes in the foreground, changing small islands of background into foreground. These techniques can also be used to find specific shapes in an image. Opening can be used to find things into which a specific structuring element can fit (edges, corners, ...).

In mathematical morphology, the closing of a set (binary image) A by a structuring element B is the erosion of the dilation of that set. In image processing, closing is, together with opening, the basic workhorse of morphological image removal. Opening removes small objects, while closing removes small holes.

PROGRAM:

```
f=imread('coins.png');
se=strel('square',20);
fo=imopen(f,se);
figure,imshow(f)
title('input image');
figure,imshow(fo)
title('opening of input image');
fc=imclose(f,se);
figure,imshow(fc)
title('opening of input image');
foc=imclose(fo,se);
figure,imshow(foc)
title('closing of opened input image');
```

Result:



Conclusion: Thus we have obtained the opened image and closed image from the original Image.

3b): AIM: To fill the region of interest for the image.

OBJECTIVE: To fill the region of interest for the image

TOOLS REQUIRED: MATLAB

THEORY: A region of interest (often abbreviated ROI), are samples within a data set identified for a particular purpose. The concept of a ROI is commonly used in many application areas. For example, in medical imaging, the boundaries of a tumor may be defined on an image or in a volume, for the purpose of measuring its size. The endocardial border may be defined on an image, perhaps during different phases of the cardiac cycle, for example, end-systole and end-diastole, for the purpose of assessing cardiac function. In geographical information systems(GIS), a ROI can be taken literally as a polygonal selection from a 2D map. In computer vision and optical character recognition, the ROI defines the borders of an object under consideration. In many applications, symbolic (textual) labels are added to a ROI, to describe its content in a compact manner. Within a ROI may lie individual points of interest (POIs).

PROGRAM:

% Program for ROI

```
clc;  
close all;  
load trees  
I=ind2gray(X,map);  
imshow(I)  
title('original image');  
I2=roifill;  
imshow(I2)  
title('OUTPUT IMAGE');
```

Result:



Original Image



Output image

Conclusion: Thus we have filled the interested region in the original image.

3c): AIM: Program for edge detection algorithm.

OBJECTIVE: Program for edge detection algorithm

TOOLS REQUIRED: MATLAB

THEORY: The Canny edge detector is an edge detection operator that uses a multi-stage algorithm to detect a wide range of edges in images. It was developed by John F. Canny in 1986. Canny also produced a computational theory of edge detection explaining why the technique works.

The Process of Canny edge detection algorithm can be broken down to 5 different steps:

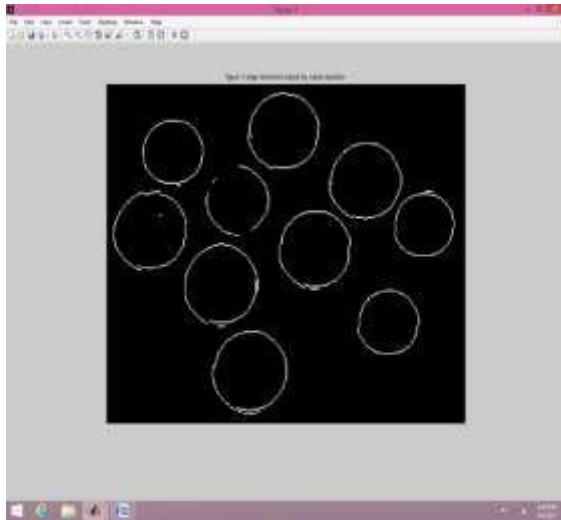
1. Apply Gaussian filter to smooth the image in order to remove the noise
2. Find the intensity gradients of the image
3. Apply non-maximum suppression to get rid of spurious response to edge detection
4. Apply double threshold to determine potential edges
5. Track edges by hypothesis: Finalize the detection of edges by suppressing all the other edges that are weak and not connected to strong edges.

PROGRAM:

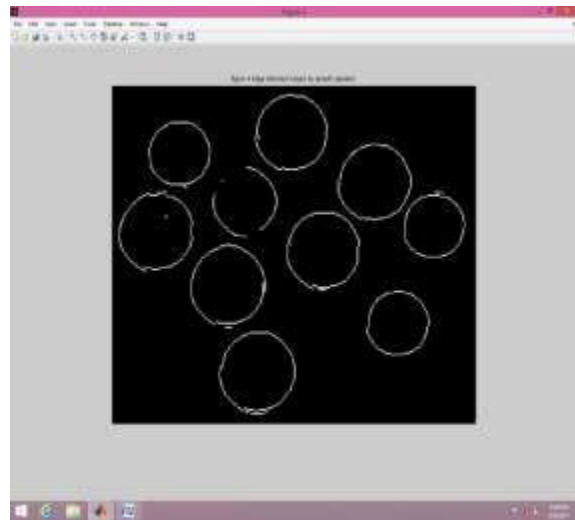
%Program for edge detection algorithm

```
I=imread('coins.png');  
figure,imshow(I)  
title('figure 1 original image');  
h=ones(5,5)/25;  
b=imfilter(I,h);  
figure,imshow(b)  
title('figure 2 filtered image');  
c=edge(b,'sobel');  
figure,imshow(c)  
title('figure 3 edge detected output by sobel operator');  
d=edge(b,'prewitt');  
figure,imshow(d)  
title('figure 4 edge detected output by prewitt operator');  
e=edge(b,'robert');  
figure,imshow(e)  
title('figure 5 edge detected output by robert operator');  
f=edge(b,'canny');  
figure,imshow(f)  
title('figure 6 edge detected output by canny operator');
```

Result:



Edge detected image



Edge detected image

Conclusion: Thus we have detected the edges in the original image.

Experiment No: 4

Date: _____

4a): Program of sharpen image using gradient mask.

AIM: Program of sharpen image using gradient mask.

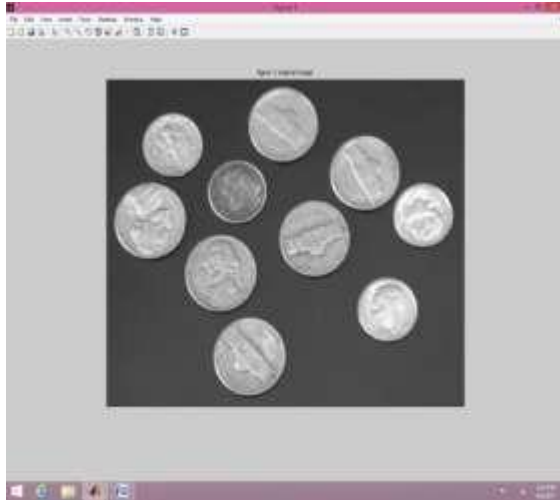
TOOLS REQUIRED: MATLAB

THEORY: An image gradient is a directional change in the intensity or color in an image. The gradient of the image is one of the fundamental building blocks in image processing. For example the Canny edge detector uses image gradient for edge detection. Mathematically, the gradient of a two-variable function (here the image intensity function) at each image point is a 2D vector with the components given by the derivatives in the horizontal and vertical directions. At each image point, the gradient vector points in the direction of largest possible intensity increase, and the length of the gradient vector corresponds to the rate of change in that direction.

PROGRAM:

```
% Program of sharpen image using gradient mask
I=imread('coins.png');
subplot(2,2,1);
imshow(I)
title('Original Image');
h=fspecial('sobel');
f=imfilter(I,h,'replicate');
subplot(2,2,2); imshow(F)
title('filtered image by sobel mask');
s=I+F;
subplot(2,2,4);
imshow(s)
title('Final o/p Image');
```

Result:



Original Image



Conclusion: Thus we have performed the sharpening operation using gradient mask on the Original image.

4b): Program for morphological operation: erosion and dilation.

AIM: Program for morphological operation: erosion and dilation.

TOOLS REQUIRED: MATLAB

THEORY: Erosion (usually represented by $\ominus$) is one of two fundamental operations (the other being dilation) in morphological image processing from which all other morphological operations are based. It was originally defined for binary images, later being extended to grayscale images, and subsequently to complete lattices.

With A and B as two sets in Z^2 (2D integer space), the dilation of A and B is defined as

$$A(+)B=\{Z|(B^{\wedge})Z\cap A\neq \phi\}$$

In the above example, A is the image while B is called a structuring element.

In the equation, $(B^{\wedge})Z$ simply means taking the reflections of B about its origin and shifting it by Z. Hence dilation of A with

B is a set of all displacements, Z, such that $(B^{\wedge})Z$ and A overlap by at least one element. Flipping of B about the origin and then moving it past image A is analogous to the convolution process. In practice flipping of B is not done always.

Dilation adds pixels to the boundaries of object in an image. The number of pixels added depends on the size and shape of the structuring element. Based on this definition, dilation can be defined as

$$A(+)B=\{\{Z|(B^{\wedge})Z\cap A\}\in A\}$$

PROGRAM:

```
% Program for morphological operations: Erosions& Dilation
```

```
f=imread('coins.png');
```

```
B=[0 1 1;1 1 1;0 1 0];
```

```
f1=imdilate(f,B);
```

```
se=strel('disk',10);
```

```
f2=imerode(f,se);
```

```
figure,imshow(f)
```

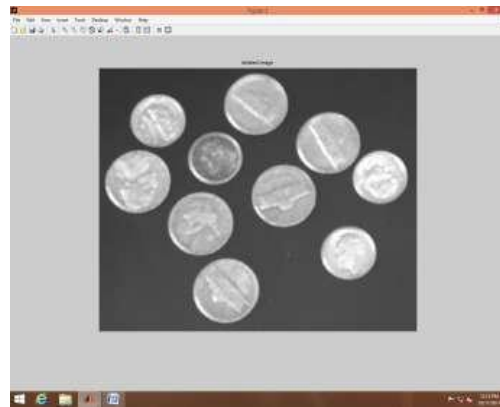
```
title('dilated image');
```

```
title('input image');
```

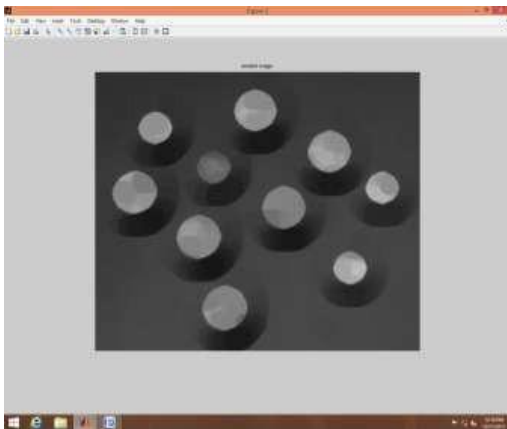


```
figure,imshow(f1)
figure,imshow(f2)
title('eroded image');
```

Result:



Eroded Image



Dilated Image

Conclusion: Thus we have obtained the eroded and dilated image for the original image

5: Program for DCT/IDCT computation.

AIM: Program for DCT/IDCT computation.

TOOLS REQUIRED: MATLAB

THEORY: A discrete cosine transform (DCT) expresses a finite sequence of data points in terms of a sum of cosine functions oscillating at different frequencies. DCTs are important to numerous applications in science and engineering, from lossy compression of audio (e.g. MP3) and images (e.g. JPEG) (where small high-frequency components can be discarded), to spectral methods for the numerical solution of partial differential equations. The use of cosine rather than sine functions is critical for compression, since it turns out (as described below) that fewer cosine functions are needed to approximate a typical signal, whereas for differential equations the cosines express a particular choice of boundary conditions.

In particular, a DCT is a Fourier-related transform similar to the discrete Fourier transform (DFT), but using only real numbers. The DCTs are generally related to Fourier Series coefficients of a periodically and symmetrically extended sequence whereas DFTs are related to Fourier Series coefficients of a periodically extended sequence. DCTs are equivalent to DFTs of roughly twice the length, operating on real data with even symmetry (since the Fourier transform of a real and even function is real and even), whereas in some variants the input and/or output data are shifted by half a sample. There are eight standard DCT variants, of which four are common.

Like for the DFT, the normalization factor in front of these transform definitions is merely a convention and differs between treatments. For example, some authors multiply the transforms by so that the inverse does not require any additional multiplicative factor. Combined with appropriate factors of $\sqrt{2/N}$, this can be used to make the transform matrix orthogonal.

PROGRAM:

```
clc;
clear all;
close all;

m=input('Enter the basis matrix dimension: ');
n=m; alpha2=ones(1,n)*sqrt(2/n); alpha2(1)=sqrt(1/n); alpha1=ones(1,m)*sqrt(2/m);
alpha(1)=sqrt(1/m); % square root. for u=0:m-1
```

```
% Request user input
```

```
for v=0:n-1
    for x=0:m-1
        for y=0:n-1
            a{u+1,v+1}(x+1,y+1)=alpha1(u+1)*alpha2(v+1)*...
            cos((2*x+1)*u*pi/(2*n))*cos((2*y+1)*v*pi/(2*n));
        end
    end
end
mag=a;
figure(3)
k=1;
```

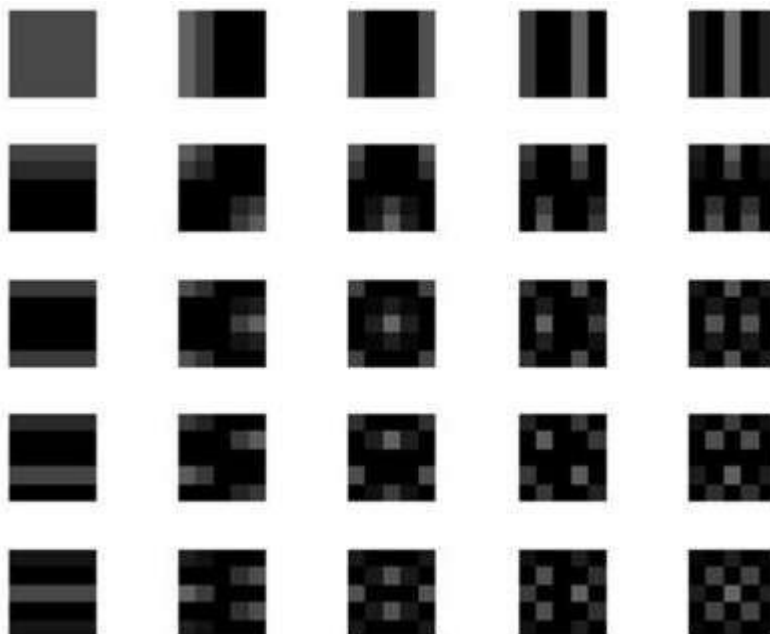
```
% Create figure graphics object
```

```
% Code to plot the basis for i=1:m
for j=1:n subplot(m,n,k)
    imshow(mag{i,j},256) k=k+1;
end end
```

Enter the basis matrix dimension: 5

Results:

```
% Create axes in tiled positions
% Display image
```



Conclusion: Thus we have obtained the DCT/IDCT for the image.

PART-B

Experiment No: 6

Date: _____

6: To read and display video file

Aim:

To read, display, process, and write a video using MATLAB.

Tools Used:

VideoReader, imshow, rgb2gray, edge, VideoWriter

Video processing is the technique of analyzing and manipulating video signals by processing individual frames over time to improve visual quality or to extract useful information. A video is a sequence of images (frames) displayed at a certain frame rate, and video processing applies image processing operations to each frame while also considering temporal (time-based) changes.

In digital signal processing and multimedia labs, video processing is typically performed using tools like MATLAB, where videos are read frame-by-frame, processed using image processing algorithms, displayed, and written back to a new video file.

6a): Reading a Video

% Reading a Video

```
Clc;  
Clear all;  
Close all;  
% Read video file  
videoObj = VideoReader('input_video.mp4');  
  
% Display video properties  
disp(videoObj)
```

6b): Displaying Video Frames

```
figure  
while hasFrame(videoObj)  
    frame = readFrame(videoObj);  
    imshow(frame)  
    title('Original Video')  
    pause(1/videoObj.FrameRate)  
End
```

Experiment No: 7

Date: _____

7: To process and write the processed video file

Aim:

To process, and write a video using MATLAB.

Tools Used:

VideoReader, imshow, rgb2gray, edge, VideoWriter

7a): Video Processing (Example: Grayscale + Edge Detection)

%Video Processing (Example: Grayscale + Edge Detection)

```
figure;
while hasFrame(videoObj)
    frame = readFrame(videoObj);

    % Convert to grayscale
    grayFrame = rgb2gray(frame);

    % Edge detection
    edgeFrame = edge(grayFrame, 'Canny');

    imshow(edgeFrame)
    title('Processed Video (Edges)')
    pause(1/videoObj.FrameRate)
end
```

7b): Writing the Processed Video to a File

```
% Create VideoWriter object
outputVideo = VideoWriter('output_video.avi');
outputVideo.FrameRate = videoObj.FrameRate;
open(outputVideo)

% Reset reader
videoObj = VideoReader('input_video.mp4');

while hasFrame(videoObj)
    frame = readFrame(videoObj);

    % Processing
    grayFrame = rgb2gray(frame);
    edgeFrame = edge(grayFrame, 'Canny');
```

```
% Convert logical to uint8 for writing
edgeFrame = uint8(edgeFrame * 255);

% Write frame
writeVideo(outputVideo, edgeFrame)
end

close(outputVideo)
disp('Video writing completed!')
```

Original - Frame 24



Blackout from 50th onwards



Blackout from 50-100 only



Waiting for frame 50...

Experiment No: 8

Date: _____

8: To read, write and process an Audio file

Aim:

To read, write and process an Audio file using MATLAB.

Tools Used:

Audioreader, Audio Writer, FFT

Program to read, write and process (plot time domain and frequency domain wave) of an Audio file

Audio processing refers to the techniques used to analyze, modify, and manipulate audio signals using digital signal processing methods. In MATLAB, audio signals can be easily read from a file, processed, and written back to a file using built-in functions.

Audio processing is widely used in:

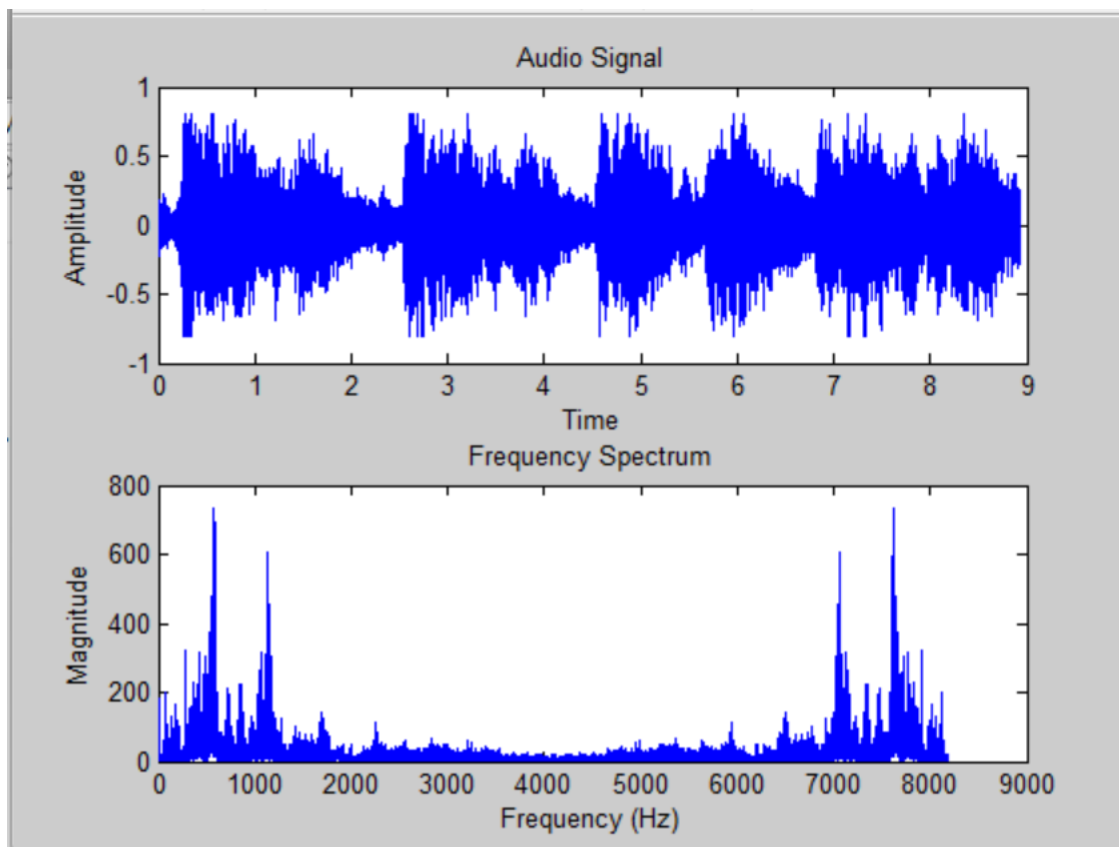
- Speech recognition systems
- Music production
- Mobile communication
- Hearing aids
- Voice assistants
- Multimedia systems

```
clc;
clear;
close all;
load handel.mat
filename = 'handel.wav';
audiowrite(filename,y,Fs);
clear y Fs
%audiowrite
[x, Fs] = audioread('handel.wav'); % Read audio file

%sound(x, Fs);           % Play audio

t = (0:length(x)-1)/Fs;
subplot(2,1,1);
plot(t, x);
title('Audio Signal');
xlabel('Time');
ylabel('Amplitude');
```

```
N = length(x);  
X = fft(x);  
f = (0:N-1)*(Fs/N);  
subplot(2,1,2);  
plot(f, abs(X));  
  
title('Frequency Spectrum');  
xlabel('Frequency (Hz)');  
ylabel('Magnitude');  
%y = flipud(x);      % Reverse audio signal  
%sound(y, Fs);
```



Experiment No: 9

Date: _____

9: Adding and removing noise for an Audio file**Aim:**

To Add noise and remove noise using filter of an Audio file using MATLAB.

Tools Used:

Audio reader, Audio Writer, filter

Noise in an audio signal refers to unwanted disturbances or random variations that degrade the quality of the original sound. Noise may occur due to environmental interference, recording equipment limitations, or transmission errors. In Audio processing, noise can be artificially added for testing and then removed using filtering techniques.

Adding noise means introducing random disturbances into the original audio signal. This is often done in MATLAB to simulate real-world noisy conditions and to test noise reduction algorithms. A common way to add noise is by using random Gaussian noise.

```
clc;
clear;
close all;
load handel.mat
filename = 'handel.wav';
audiowrite(filename,y,Fs);
clear y Fs
% Step 1: Read audio file
[x, Fs] = audioread('handel.wav');

% Time axis
t = (0:length(x)-1)/Fs;

% Step 2: Generate random noise
noise = 0.1 * randn(size(x));

% Step 3: Add noise to audio signal
noisy_signal = x + noise;

% Step 4: Display original and noisy signals
figure;

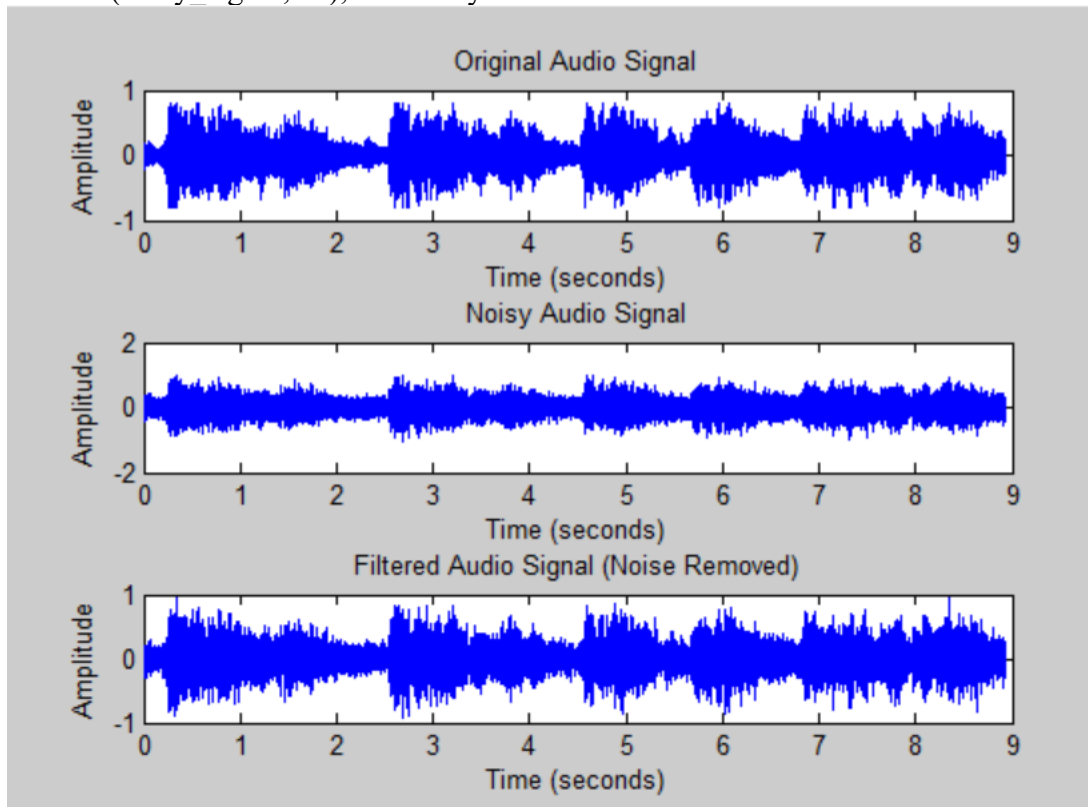
subplot(3,1,1);
plot(t, x);
title('Original Audio Signal');
xlabel('Time (seconds)');
ylabel('Amplitude');
```

```
subplot(3,1,2);
plot(t, noisy_signal);
title('Noisy Audio Signal');
xlabel('Time (seconds)');
ylabel('Amplitude');

% Step 5: Remove noise using Low Pass Filter
fc = 1500; % Cutoff frequency
[b,a] = butter(6, fc/(Fs/2)); % (Butterworth filter(6 -order of the filter,fc/FS/2 - Normalized
% cutoff frequency)
filtered_signal = filter(b, a, noisy_signal); % (a, b are filter coefficients)

% Step 6: Display filtered signal
%figure;
subplot(3,1,3);
plot(t, filtered_signal);
title('Filtered Audio Signal (Noise Removed)');
xlabel('Time (seconds)');
ylabel('Amplitude');

% Play signals
%sound(x, Fs); % Original
%pause(3);
%sound(noisy_signal, Fs); % Noisy
```



PROGRAM OUTCOMES (POs)

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.